

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnaury W.		
Certified By	09-Jun-2012	Mr.Amnaury W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnaury W.	Issue for Approved
1	13-Feb-2013	Mr.Chatchawan S.	Mr.Phuchin S.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-004-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-006-1**



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. **RNC-WPS-006-1** Date **13-Feb-13** Revision No. **1**

Supporting PQR No.(s) **RNC-PQR-006-1** Date **13-Feb-13** Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding + Shield Metal Arc welding**
(GTAW+SMAW)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal

☐ Nonmetallic ☐ Other

Notes _____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1,2** to P-No. **1** Group No. **1,2**

Material Specification Type and Grade **SA 333 Gr.1,6 SA420 WPL6** TO **SA 333 Gr.1,6 SA420WPL6**

Chem. Analysis and Mech. Prop. _____

Thickness Range :

Base Metal : Groove **1.6 mm - 14.22 mm.** Fillet **All**

Pipe Dia. Range : Groove **All** Fillet **All**

Other _____

FILLER METALS (QW-404)

	GTAW	SMAW
Spec. No. (SFA)	5.28	5.5
AWS No. (Class)	ER-70S-G	E8016-G
F-No.	6	4
A-No.	1	1
Size of Filler Metals	Ø 2.0 & 2.4 mm	Ø 3.2 mm.
Deposited Weld Metals	2.0 mm	5.11 mm.
Thickness Range :		
Groove	4 mm Max.	10.22 mm. Max.
Fillet	All	All
Electrode-Flux (Class)	-	-
Flux Trade Name	-	-
Consumable Insert	-	-
Other	Kobelco TGS- 1N or Equivalents	Kobelco NB- 1 or Equivalents

POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression Up ☒ X Down ☐

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **N/A**

Time Range **N/A**

Notes _____

PREHEAT (QW-406)

Preheat Temp. Min. **10°C**

Interpass Temp. Max. **238°C**

Preheat Maintenance **N/A**

Notes _____

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	13-17Ltr/Min
Trailing	-	-	-
Backing	-	-	-

WPS No. RNC-WPS-006-1 Rev. 1

ELECTRICAL CHARACTERISTICS (QW-409)

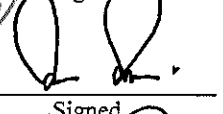
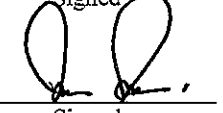
Current ☒ AC ☒ DC Polarity EN(GTAW)+EP(SMAW)
 Amps (Range) 90-140(GTAW) + 80-130(SMAW) Volts (Range) 9-13(GTAW) + 18-24(SMAW)
 Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

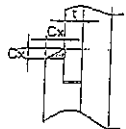
String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 6.0-10.0 Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 70S-G	2.4	DC EN	90-130	9-12	6.0-10.0	
2nd	GTAW	ER 70S-G	2.4	DC EN	100-140	10-13	6.0-10.0	
3 & Over	SMAW	E8016-G	3.2	AC/DCEP	80-130	18-24	6.0-10.0	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of ASME-IX, 2007 latest Revision.

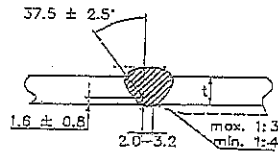
Prepared by: Mr. Chatchawan  13/03/13
 Name Signed Date
 Reviewed by: ภูชิน สีหอมไธ  22/03/13
 Name Signed Date
 Approved by: ภูชิน สีหอมไธ  22/03/13
 Name Signed Date

ATTACHED FIGURE JOINT SKETCH



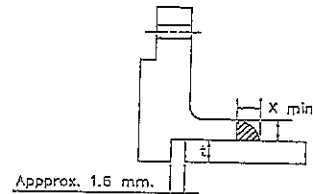
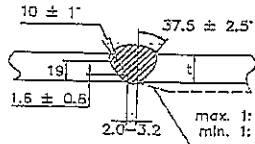
Cx (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



Misalignment (max.) = 1.5 mm.

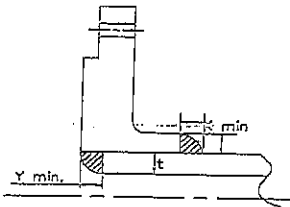
JOINT No. B1 & B2



Approx. 1.5 mm.

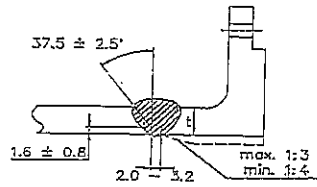
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



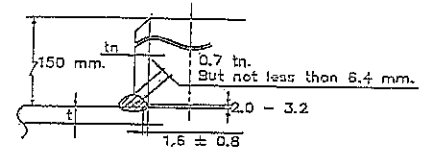
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



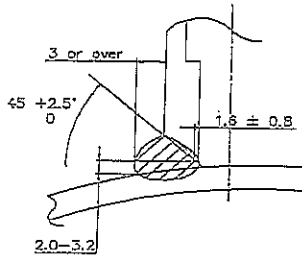
Misalignment (max.) = 1.5 mm.

JOINT No. B3

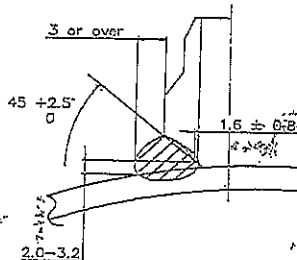


Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

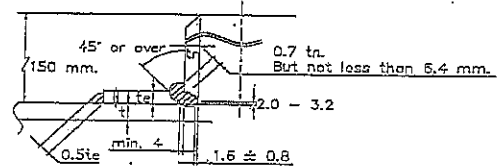
JOINT No. B4



JOINT No. C1

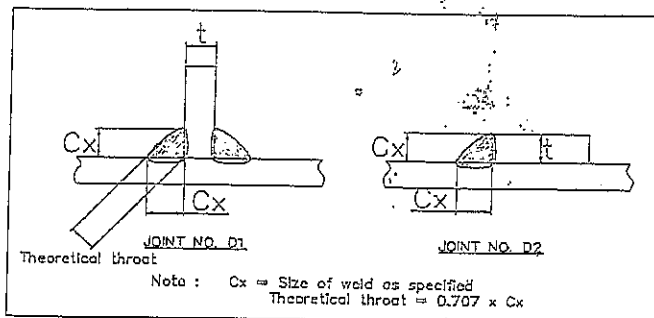


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

JOINT No. B5



JOINT No. D1

JOINT No. D2

Note : Cx = Size of weld as specified
Theoretical throat = $0.707 \times Cx$

Procedure Qualification Record (PQR)

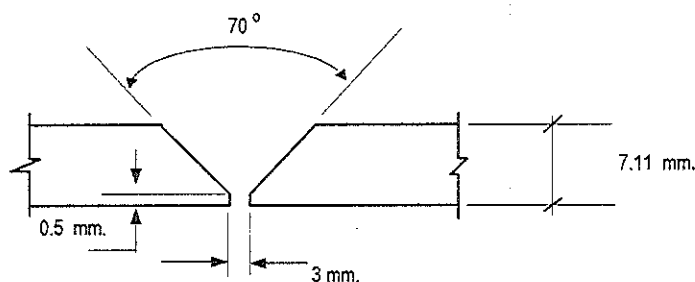
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-006-1

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-006-1	Date :	13 February 2013
WPS Number :	RNC-WPS-006-1		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	A333	To	A333	Temperature	N/A		
Type or Grade	Gr.6	To	Gr.6	Hold Time	N/A		
P. No.	1	To	1	Others			
Thickness of Test Coupon		7.11 mm.		GAS (QW-408)			
Diameter of Test Coupon		Ø 6"					
Others							
FILLER METALS (QW-404)				Percent Composition			
					Gas (es)	Mixture	Flow Rate
				Shielding	Argon	99.99%	20 L/Min.
				Trailing	-	-	-
				Backing	-	-	-
SFA Specification		GTAW	SMAW	ELECTRICAL CHARACTERISTICS (QW-409)			
AWS Classification		A 5.28	A 5.5	Current	DC (GTAW)	DC (SMAW)	
Filler Metal F-No.		ER70S-G	E8016-G	Polarity	EN	EP	
Weld Metal Analysis A-No.		6	4	Amperage	102 - 110 (GTAW)	Voltage	10 - 10 (GTAW)
Size of Filler Metal		1	1	95 - 98 (SMAW)		23 - 23 (SMAW)	
Others		Ø 2.4 mm.	Ø 3.2 mm.	Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2		
Weld Metal Thickness		Kobelco TGS-1N	Kobelco NB-1	Others	N/A		
POSITION (QW-405)				TECHNIQUE (QW-410)			
Position of Groove		6G		Travel Speed	4.36 - 9.49 cm/min.		
Progression of Welding (Uphill or Down)		Up Hill		String or Weave Bead	Both		
Others		N/A		Oscillation	N/A		
PREHEAT (QW-406)				Multiple or single Pass		Multiple Pass	
Preheat Temperature		35° C		Multiple or Single Electrodes		Multiple Electrodes	
Interpass Temperature (max.)		177° C		Others		N/A	
Others		N/A					

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- RNC-PQR-006-1

Tensile Test Report No.TS-13-02-087

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
548/13 TR1	19.23	6.65	127.88	59.73	467.09	OUT OF WELD
548/13 TR2	18.99	6.92	131.41	61.62	468.90	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-02-036

Type and Figure Number	Type of Bend	Indication	Results
548/13 BF1	Transverse face bend	1.57,0.91 mm. (WELD ZONE)	Accepted
548/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
548/13 BR1	Transverse root bend	No Open Discontinuity	Accepted
548/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.WQT-007

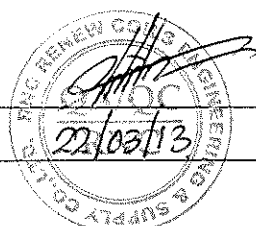
Hardness Testing Report No. HV-13-02-021

Macrostructure Report No. MA-13-02-055

Impact Testing Report No.IM-13-03-006 (-60°C)

Welders Name Mr.Pongpet H. Welder Number W-002
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

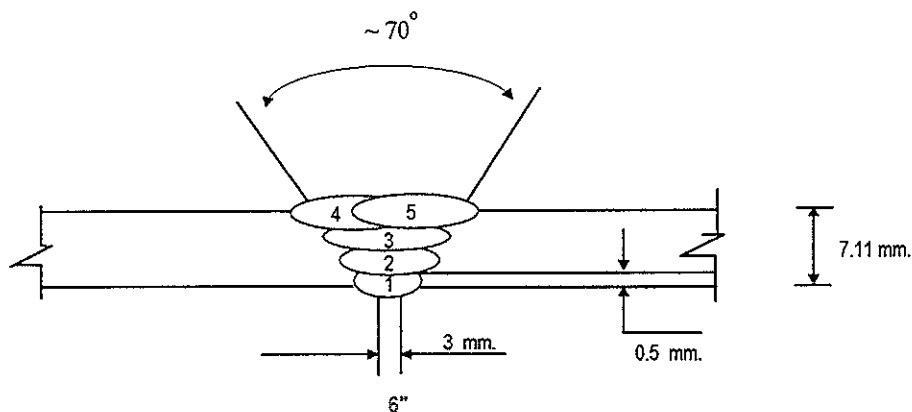
We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor : 
Date : 22/03/13

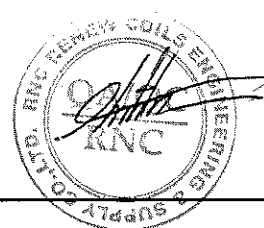
Inspected by : Mr.Wichit K.
Certified by : Mr.Chaloemkiet R.
Date : 13 March 2013 

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-006-1
		WPS No. : RNC-WPS-006-1
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST
		CONDITION AND RESULT

Welding Sequences for RNC-PQR-006-1 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER70S-G	2.4	102	10	DCEN	-	-	4.36	14.04	20	-
2	GTAW	ER70S-G	2.4	110	10	DCEN	-	-	5.82	11.34	20	-
3	SMAW	E8016-G	3.2	95	23	DCEP	-	-	7.75	16.92	-	-
4	SMAW	E8016-G	3.2	96	23	DCEP	-	-	7.39	17.93	-	-
5	SMAW	E8016-G	3.2	98	23	DCEP	-	-	9.49	14.25	-	-





PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT
 69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
 Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

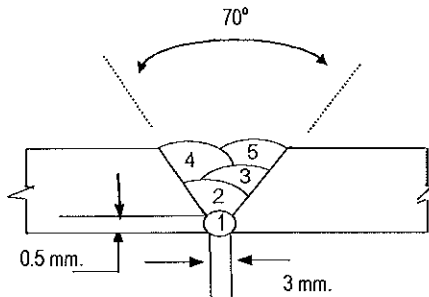
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-016	
		Customer Order No. : N/A		WPS No. : RNC-WPS-006-1		Date : 13-Feb-13	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-002	MR.PONGPET H.	GTAW / SMAW	6G	ASME IX	A333Gr.6	ø 6" Thk. 7.11 mm.	A5.28 / A5.5	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	DCEN	102	10	4.36	35	14.04	Argon	20	-	-
2	GTAW	ER70S-G	2.4	DCEN	110	10	5.82	169	11.34	Argon	20	-	-
3	SMAW	E8016-G	3.2	DCEP	95	23	7.75	154	16.92	-	-	-	-
4	SMAW	E8016-G	3.2	DCEP	96	23	7.39	170	17.93	-	-	-	-
5	SMAW	E8016-G	3.2	DCEP	98	23	9.49	177	14.25	-	-	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

Mr.Wichit K.

14-Feb-2013

SIGNATURE :

NAME :

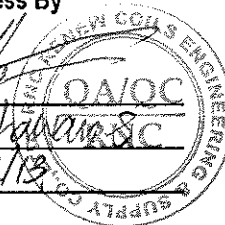
DATE :

Mr.Chatchawan S.
19/02/13

SIGNATURE :

NAME :

DATE :



PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229

Page : 1 Of 1

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnmpm.co.th

29783

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-02-087

Revision No. 0

Test Product	WPS No. RNC-WPS-006-1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Test	Reduced Section Tension
Dimension	Ø 6" (7.11 mm Thk.)	Equipment's Capacity	100 TONS
Date Received	19 February 2013	Equipment / Serial No.	INSTRON / 1000DXR1414
Date Tested	20 February 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	548/13 TR1	N/A	6.65	19.23	127.88	N/A	N/A	N/A	59.73	467.09	N/A	OUT OF WELD
2	548/13 TR2	N/A	6.92	18.99	131.41	N/A	N/A	N/A	61.62	468.90	N/A	OUT OF WELD

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Pongpet H.

ID No. 3 2512 00585 27 8

Welder No. : W-002

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th

29789

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-02-036

Revision No. 0

Test Product	WPS No. RNC-WPS-006-1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 6" (7.11 mm Thk.)	Diameter of Meudrel	26 mm
Date Received	19 February 2013	Bend Angle	180 Degree.
Date Tested	20 February 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	548/13 BF1	Transverse face bend	6.40x37.85x162.49 mm	Open Discontinuity	WELD ZONE	1.57, 0.91 mm
2	548/13 BF2	Transverse face bend	6.89x38.01x162.09 mm	No Open Discontinuity	N/A	-
3	548/13 BR1	Transverse root bend	6.83x37.96x162.20 mm	No Open Discontinuity	N/A	-
4	548/13 BR2	Transverse root bend	6.94x37.87x162.20 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Pongpet H.

ID No. 3 2512 00585 27 8

Welder No. : W-002

Reported by :

(Mr. Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by :

(Mr. Pavaret Freedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasangutjeree, Samutprakarn 10290 Thailand.

13235

LPN PLATE MILL

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Report No. HV-13-02-021

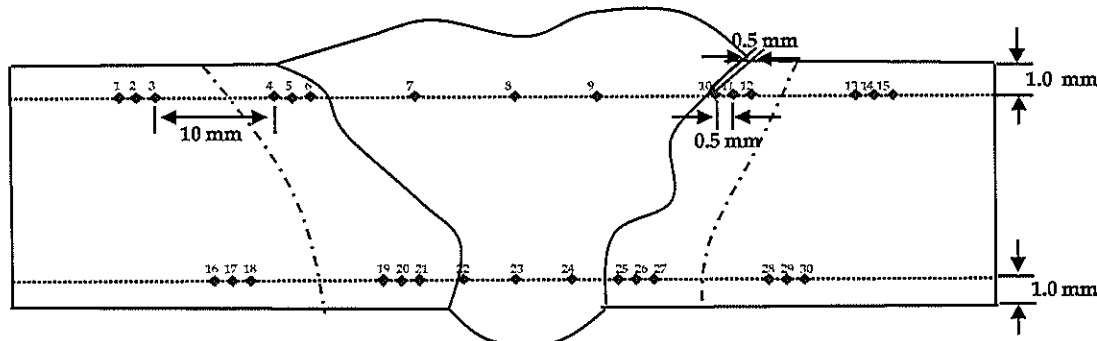
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.

Revision No. 0

Test Product	WPS No. RNC-WPS-006-1	Material Characterization	Pipe
Test No.	548/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ⁶²
Project Name	WELDER SUPPLY AND PQR	Standard of	ASME Section IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A333 Gr.6	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	6" (7.11 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW+SMAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	18 February 2013	Cal. Block	No. A60023 No. A62072
Date Tested	20 February 2013	Ambient Temperature	26.0 °C

Test Location

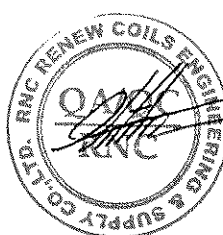
Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	130.8	16	BASE	131.0			*****
2	BASE	131.7	17	BASE	130.6			
3	BASE	131.4	18	BASE	131.1			
4	HAZ	154.9	19	HAZ	145.4			
5	HAZ	178.6	20	HAZ	148.1			
6	HAZ	183.7	21	HAZ	156.5			
7	WELD	196.4	22	WELD	149.6			
8	WELD	194.5	23	WELD	148.8			
9	WELD	201.7	24	WELD	148.2			
10	HAZ	179.3	25	HAZ	153.3			
11	HAZ	162.1	26	HAZ	149.8			
12	HAZ	152.0	27	HAZ	148.8			
13	BASE	130.1	28	BASE	130.7			
14	BASE	130.3	29	BASE	129.8			
15	BASE	131.4	30	BASE	131.1			

Additional details :-

Reported by :
(Mr. Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22/02/13



Approved by :
(Mr. Pavaret Preedawiphat)
Deputy Managing Director
Date : 22/02/13

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Testing Laboratory LPN Plate Mill Public Company Limited

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13699

Macrostructure Analysis Report

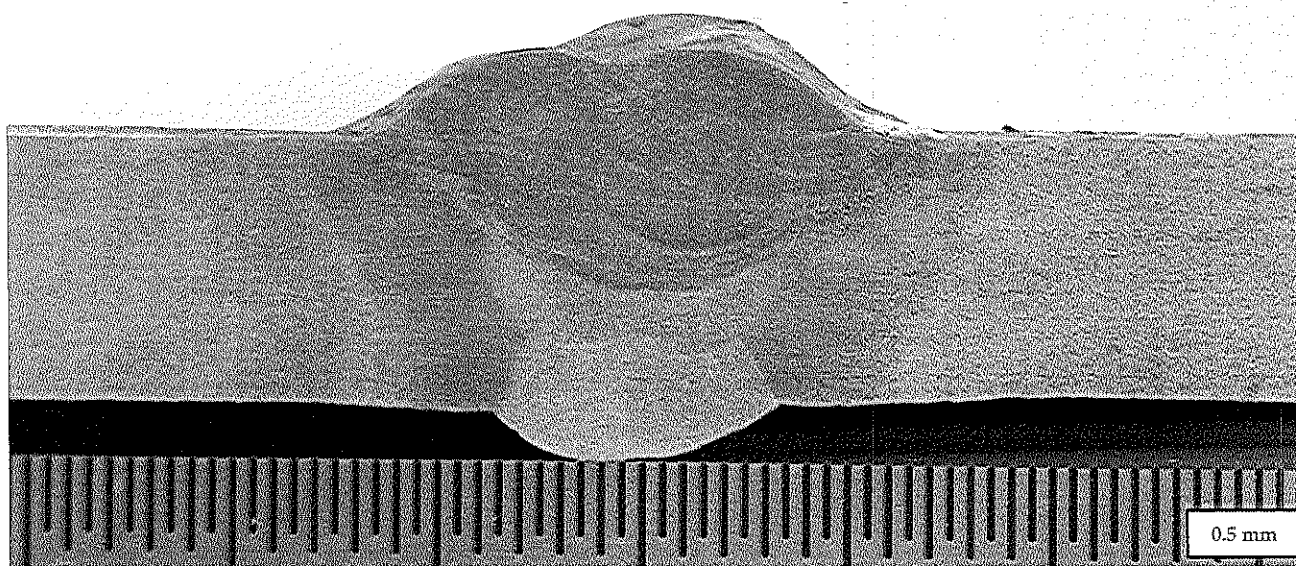
Page 1 of 1

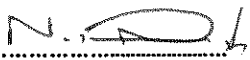
Report No. MA-13-02-055

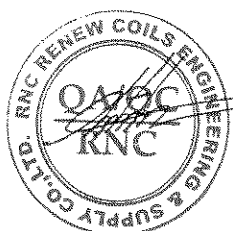
Revision No.0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.**Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.**

Test Product	WPS No. RNC-WPS-006-1	Material Characterization	Pipe
Test No.	548/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A333 Gr.6	Method of Preparation	ASME Section IX : 2010 edition
Dimension	6" (7.11 mm Thk.)	Test Method	ASME Section IX : 2010 edition
Welding Process/Position	GTAW+SMAW/6G	Standard of Test	ASME Section IX : 2010 edition
Type of Joint	Butt Jiont	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	Mr. Pongpet H. / W-002	Time of Etching	5 minutes
Date Received	18 February 2013	Date Tested	20 February 2013

**Macro-etch Examination :** The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.**Additional details :** -

Reported by : 
 (Mr.Keerati Sa-ngoen)
 (Acting) General Manager - Quality Assurance
 Date : 22/02/13



Approved by : 
 (Mr.Pavaret Preedawiphat)
 Deputy Managing Director
 Date: 22/02/13

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th

28609

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-03-006

Revision No. 0

Test Product	WPS No. RNC-WPS-006-1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 and ASTM E23M-07a ^{e1}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Test	Charpy V-Notch Impact
Dimension	Ø 6" (7.11 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Received Date	28 February 2013	Equipment 's Capacity	542 Joules
Tested Date	04 March 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	548/13-1	-60°C	5.016	10.011	54.91	125.05	119.02	Base Metal	-
2	548/13-2	-60°C	5.011	10.013	54.91	114.28			
3	548/13-3	-60°C	5.015	10.009	54.92	117.73			
4	548/13-1	-60°C	5.015	10.009	54.95	119.41	119.93	HAZ	-
5	548/13-2	-60°C	5.013	10.006	54.91	116.90			
6	548/13-3	-60°C	5.008	10.009	54.85	123.48			
7	548/13-1	-60°C	5.015	10.009	54.92	110.48	104.99	Weld Metal	-
8	548/13-2	-60°C	5.012	10.008	54.86	106.70			
9	548/13-3	-60°C	5.014	10.007	54.89	97.79			

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Pongpet H.

ID No. 3 2512 00585 27 8

Welder No. : W-002

Reported by :

(Mr. Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

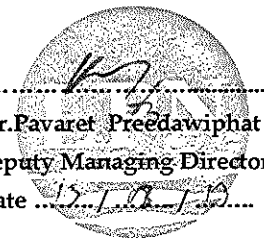
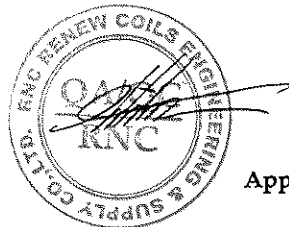
Date / /

Approved by :

(Mr. Pavaret Preedawiphat)

Deputy Managing Director

Date / /

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